

Raúl Quibén Figueroa

Optimization, trajectory design & data-driven modeling for sustainable aviation

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PROFILE

Aerospace engineer finishing a PhD on the **design and trajectory optimization of hydrogen-powered aircraft** at Universidad Carlos III de Madrid. I turn open engineering questions into well-posed optimization problems and solve them with my own Python tools: multidisciplinary design optimization, optimal control (direct and indirect methods) and large-scale data analysis of flights, weather and contrails. My work is peer-reviewed and won a national innovation award, and I have researched in Spain, Italy, the US (Stanford) and Japan (Tohoku).

EXPERIENCE

Jul 2022 – present	Predoctoral Research Fellow · Aircraft Operations Lab, Dept. of Aerospace Engineering, UC3M - Built MOTIVATION, a multidisciplinary design-optimization framework in Python (OpenMDAO, pyOptSparse) for hydrogen-powered and hybrid-electric transport aircraft, integrating aircraft design, performance, propulsion, cost and emissions models. - Formulated and solved large-scale optimal-control problems: an indirect (PMP) solver for optimal cruise of LH₂ aircraft and a direct 4D trajectory-optimization campaign over meteorological data, including contrail (non-CO₂) climate effects. - Funded by the HYDROGENATING-CM-UC3M project (2022–23) and a competitive Community of Madrid predoctoral contract (2023–26); contributed to proposals for the SESAR project ZENITIA on holistic climate-impact mitigation. - Published in JCR journals and presented at AIAA AVIATION, ECCOMAS AeroBest, EGU and ECATS; first-author paper awarded the Luis Azcárraga Aeronautic Innovation Award 2025.
May – Jul 2024	Visiting Researcher · Stanford University, Civil & Environmental Engineering (Prof. C. Gorlé) Compared high-fidelity large-eddy simulations of early contrail formation with ground-based all-sky camera observations, together with Dr. T. Ferreira. Funded by a UC3M mobility grant; results presented at EGU 2025.
Feb – Jun 2020	Research Collaborator · University of Vigo (DRONEIoT project) Programmed and tested a LoRa-based drone communication system.
May – Jul 2019	Engineering Intern · AIMEN Technology Centre, Calculation & Simulation Dept. Finite-element analysis in Abaqus for structural calculation, certification and project delivery.

EDUCATION

2022 – present	PhD in Aerospace Engineering · Universidad Carlos III de Madrid <i>H₂-fuelled aircraft configuration and trajectory optimization to reduce aeronautic environmental impact.</i> Supervisors: Dr. Manuel Soler and Dr. Rauno Cavallaro.
2020 – 2022	MSc Aeronautical Engineering · Universidad Carlos III de Madrid Qualifying degree taught in English (7.89/10). Thesis: platform for the design and optimization of an H ₂ -powered regional aircraft. Erasmus+ semester at Politecnico di Milano (2022).
2016 – 2020	BSc Aerospace Engineering · Universidade de Vigo 8.42/10, top 10% of the class; 2nd best national graduate (SEDEA 2020). Bachelor thesis at Tohoku University, Japan (Space Structures Lab, Prof. K. Makiyara); semi-active vibration control with piezoelectric transducers.

PUBLICATIONS

SELECTED

JOURNAL	Feasibility studies on regional aircraft retrofitted with hybrid-electric powertrains Award 2025 R. Quibén Figueroa, R. Cavallaro, A. Cini · <i>Aerospace Science and Technology</i> 151, 109246 (2024) · doi:10.1016/j.ast.2024.109246
JOURNAL	Multidisciplinary design optimization of hydrogen-powered transport aircraft R. Quibén Figueroa, R. Cavallaro, M. Soler · <i>Journal of Aircraft</i> , under review
JOURNAL	Optimal cruise of hydrogen transport aircraft R. Quibén Figueroa, A. Jafarimoghaddam, M. Soler, R. Cavallaro · <i>Aerospace Science and Technology</i> , submitted
CONFERENCE	MOTIVATION (Mdao fOr susTaInable aViATION): framework for the design and optimization of H₂-powered aircraft AIAA AVIATION 2023 Forum, San Diego · doi:10.2514/6.2023-4151
CONFERENCE	Multidisciplinary design and optimization of H₂-powered regional aircraft architectures P. Norczyk Simon, R. Quibén Figueroa, A. Cini, R. Cavallaro · AIAA SciTech 2025, Orlando · doi:10.2514/6.2025-0360
CONFERENCE	Hydrogen-powered aircraft design: a multidisciplinary approach to minimize emissions and cost AeroBest 2025, ECCOMAS Thematic Conference on MDO of Aerospace Systems, Lisbon
TALK	Comparing high-fidelity LES of early contrail formation with ground-based images EGU General Assembly 2025, Vienna · doi:10.5194/egusphere-egu25-17191
POSTER	Toward climate-optimal trajectory optimization for H₂-powered aircraft 5th ECATS Conference, Brussels, 2026
TALK	Multidisciplinary optimization of H₂-powered aircraft to reduce emissions and costs 4th ECATS Conference, Delft, 2023

Awards & Grants

- **Luis Azcárraga Aeronautic Innovation Award 2025**, Fundación ENAIRE (€10,000)
 - **Community of Madrid predoctoral contract 2023**, competitive call (score 8.98)
 - **UC3M Mobility Grant 2024** for the Stanford research stay
 - **2nd Best National Graduate** in Aerospace Engineering, SEDEA 2020
- **JASSO Excellence Scholarship**, Japanese Government (2019), and Ge4 scholarship
 - **Academic Excellence Awards**, Xunta de Galicia (2020) and Universidade de Vigo (2017)
 - Ministry of Education collaboration grant (2019) · Erasmus+ (2022)
 - Extraordinary Baccalaureate Award, Xunta de Galicia (2016)

Teaching, Training & Outreach

- Flight-simulator laboratory sessions for BSc and MSc students, UC3M (2024 – 2026)
 - Guest lecture on hydrogen-powered aircraft design, aircraft design course, UC3M (2024)
 - Flight-plan assignments, Air Navigation Systems, UC3M (2024/25)
 - Outreach: Madrid Science Fair 2024 · IES Pozuelo 2024
- NATO AVT-385 Lecture Series on multi-fidelity MDO, Rome (2024)
 - ECATS More&Less Autumn School on aviation climate impact, Munich (2023)
 - SESAR Innovation Days, Seville (2023)
 - HPC in heterogeneous architectures · ParaView · GitHub & Copilot (UC3M)

Skills

Optimization

MDO · Optimal control · PMP · Direct methods · Gradient-based NLP · SNOPT

Programming & data

Python · NumPy · SciPy · pandas · OpenMDAO · pyOptSparse · CasADi · MATLAB · Git · HPC

LaTeX

Aviation & climate

Aircraft design & performance · Propulsion · LH₂ systems · ATM · pycontrails · CoCiP

ADS-B / OpenSky

Simulation

FEM (Abaqus) · OpenVSP · ParaView · Computer vision (OpenCV)

Languages

Spanish native
Galician native
English C1 · Cambridge CAE
Japanese A1
Portuguese · Italian basic